Mandhagini Madhavan

mandhagini23@gmail.com +44 7587 583053 United Kingdom LinkedIn GitHub Portfolio

PROFESSIONAL SUMMARY

MSc Artificial Intelligence graduate with hands-on experience developing AI applications using **Python, R, FastAPI, React and LLM APIs**. Built an AI investigation platform incorporating **Retrieval-Augmented Generation (RAG), multi-agent workflows and evidence-based analysis**. Skilled in machine learning, NLP and SQL, with practical experience in REST API development, model integration and automated testing.

TECHNICAL SKILLS

Programming: Python, SQL, R, JavaScript, TypeScript
AI & Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent AI Systems, Prompt Engineering
Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, FastAPI, Streamlit, React, OpenCV
AI & Backend Engineering: REST APIs, Semantic Search, Vector Embeddings, Model Integration, Automated Testing
Data Analysis & Visualisation: pandas, NumPy, Data Cleaning, Exploratory Data Analysis (EDA), Matplotlib, Excel (Pivot Tables, VLOOKUP), Power BI
Cloud & Development Tools: AWS (S3, EC2), Docker, Git, GitHub, pytest, FFmpeg, Jupyter Notebook

PROJECT EXPERIENCE

InsightForge Multi-Agent AI Investigation Platform
- Developed a full-stack AI investigation prototype using **Python, FastAPI, React and TypeScript**, with REST API endpoints and a modular architecture.
- Designed an architecture for **RAG, investigator–critic agent workflows, evidence retrieval and human review**, supporting traceable AI-generated investigations.

TimeSync AI — Intelligent Timestamp Reconciliation & Video Clipping
- Engineered a **Python-based timestamp reconciliation system** using confidence-based decision logic to resolve inconsistencies between video and speech transcription timestamps.
- Integrated **FFmpeg** for video clipping and validated reconciliation logic through **22 passing automated tests using pytest**.

Persona Multiplier AI - LLM-Powered Content Generation
- Built an interactive **Streamlit application** using LLM APIs to transform a single source story into **three recruiter-focused content styles**.
- Applied **prompt engineering and persona-specific generation** to produce tailored professional messaging.

Wildfire Detection System — Computer Vision & NLP
- Developed a multimodal research prototype using **CNN, MobileNetV2, ResNet50 and Attention CNN** for wildfire image classification.
- Integrated image predictions with **TF-IDF and Logistic Regression sentiment analysis** to demonstrate rule-based wildfire alert generation.

PROFESSIONAL EXPERIENCE

Independent AI & Software Developer, Personal Projects 2025 – Present
- Designed and developed AI applications using **Python, FastAPI, React, TypeScript and LLM APIs**, focusing on practical AI integration and software development.
- Applied **REST API development, model integration, automated testing and Git-based version control** across independent technical projects.

Clerical Marker, Cambridge University Press & Assessment 05/2024 – 08/2024 | United Kingdom
- Reviewed and validated high volumes of assessment data, maintaining accuracy and consistency against established quality standards.
- Worked to strict deadlines, demonstrating attention to detail, data validation and effective collaboration.

AI & Data Science Intern, Freelance 01/2023 – 08/2023 | India
- Developed Python-based machine learning and data analysis projects involving data preprocessing, model development and performance evaluation.
- Applied data cleaning and exploratory analysis techniques to structured datasets.

EDUCATION

MSc Artificial Intelligence, Anglia Ruskin University United Kingdom (Completed: 2025)

Bachelor of Computer Applications (BCA), Bishop Heber College India (Completed: 2023)

CERTIFICATIONS

AWS Academy Machine Learning Foundations

LANGUAGES

Tamil — Native/Bilingual **English** — Fluent